

Climate Health Vulnerability Cluster Analysis Report

Rochester, NY Census Tracts
Analytics Division

Executive Summary

The analysis identifies **4 distinct vulnerability clusters** among Rochester, NY census tracts. **Cluster 3** ($n = 12$ tracts) is the highest-risk group, with the greatest poverty burden, strongest racial and ethnic concentration, and most severe housing and access barriers. **Cluster 1** ($n = 66$ tracts) represents the lowest vulnerability baseline.

1 Results and Interpretation

1.1 Clustered Vulnerability Patterns

The all-cluster heatmap shows clear separation in vulnerability indicators across neighborhoods. High values are concentrated in a limited group of tracts, suggesting spatial clustering of disadvantage rather than a citywide uniform pattern. Cluster 3 stands out as the most vulnerable group, with elevated poverty, housing stress, limited English proficiency, and lower life expectancy. Cluster 1 appears comparatively advantaged, with lower poverty and much lower transit dependence.

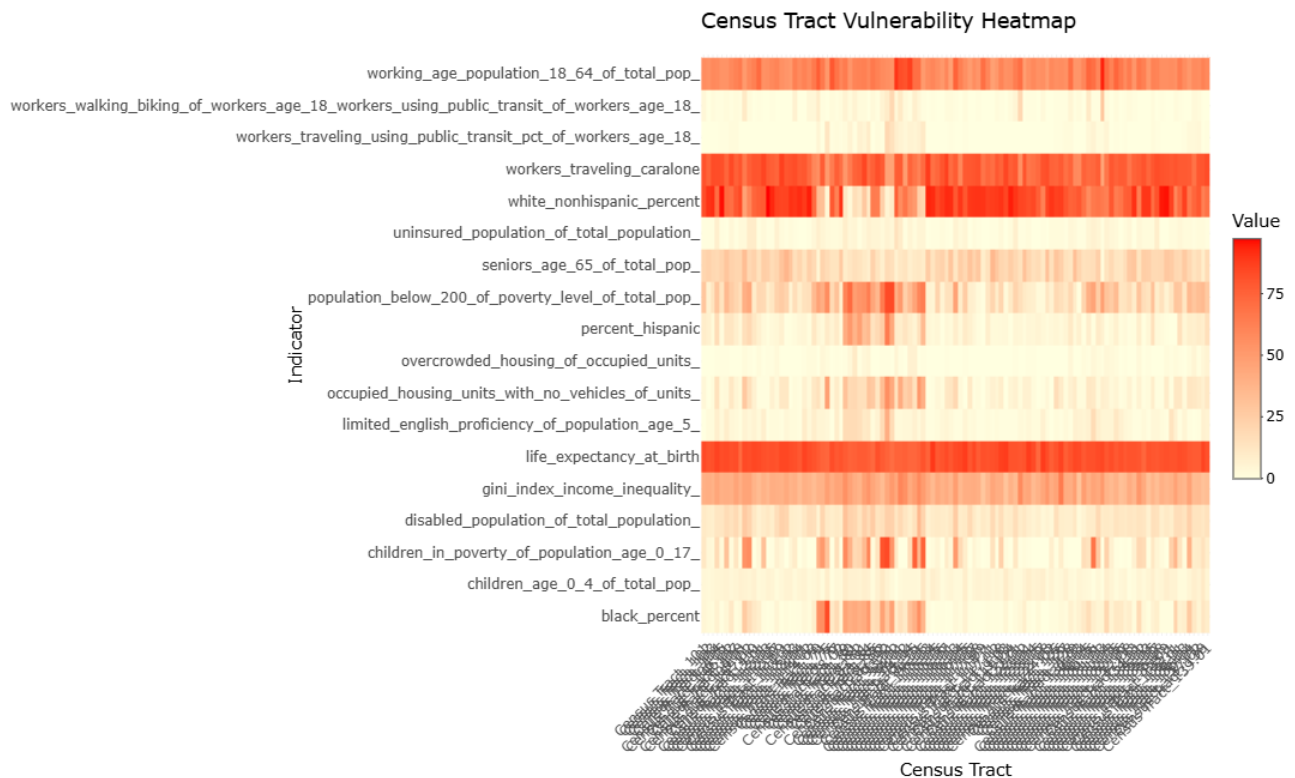


Figure 1: Heatmap of all census tracts showing the distribution of vulnerability indicators across clusters.

1.2 PCA and Cluster Separation

The PCA scatterplot shows that the four clusters are separated in reduced feature space. Cluster 1 is concentrated on the left side of the plot, Cluster 2 lies in the middle, Cluster 3 is shifted to the right and upward, and Cluster 4 forms a smaller isolated group. This pattern confirms that the clustering captures meaningful structural differences across census tracts.

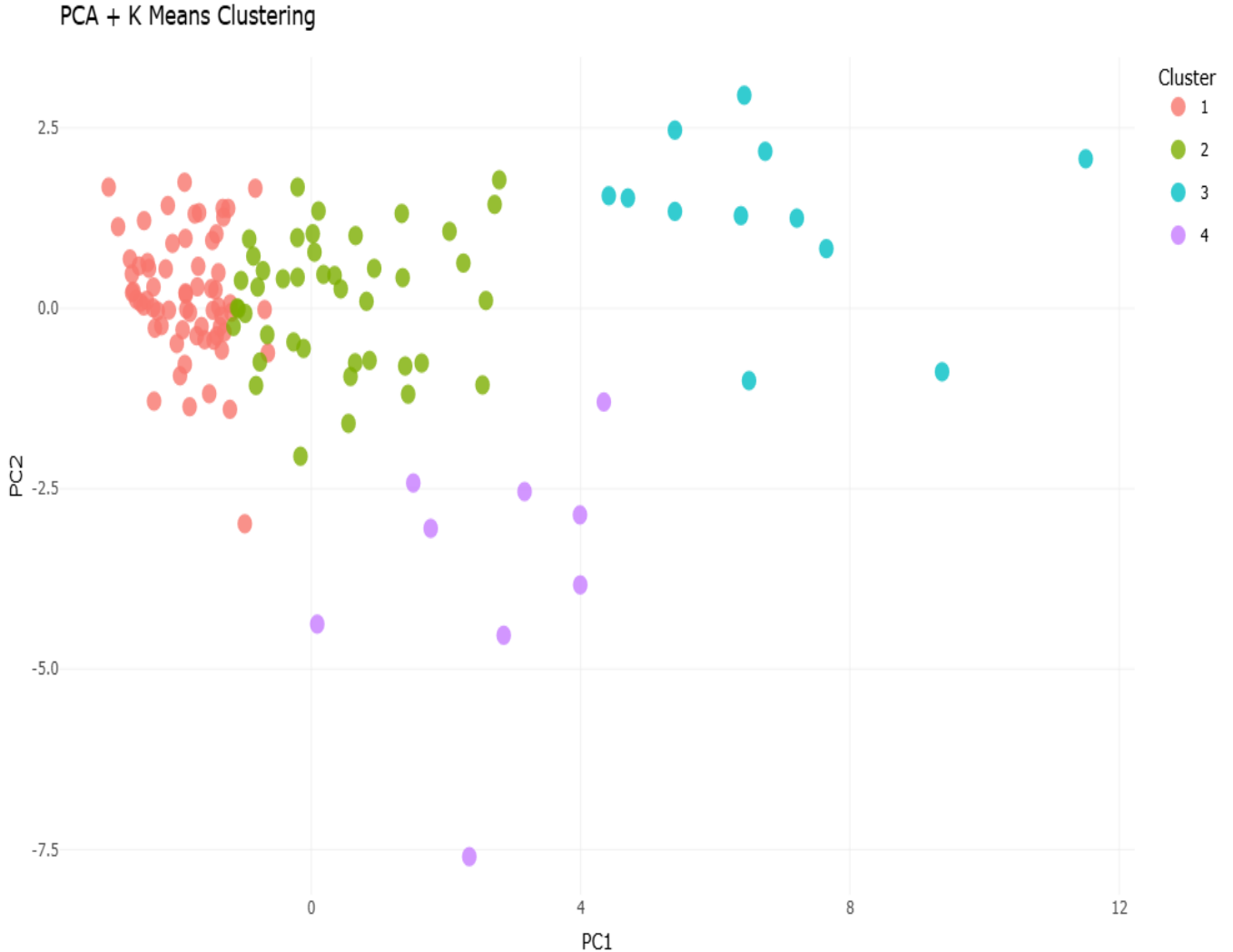
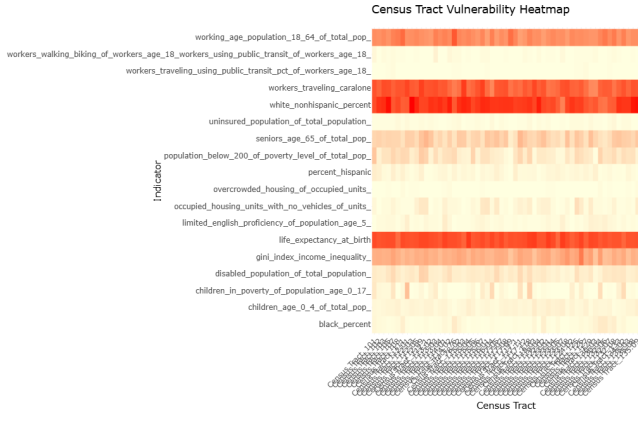


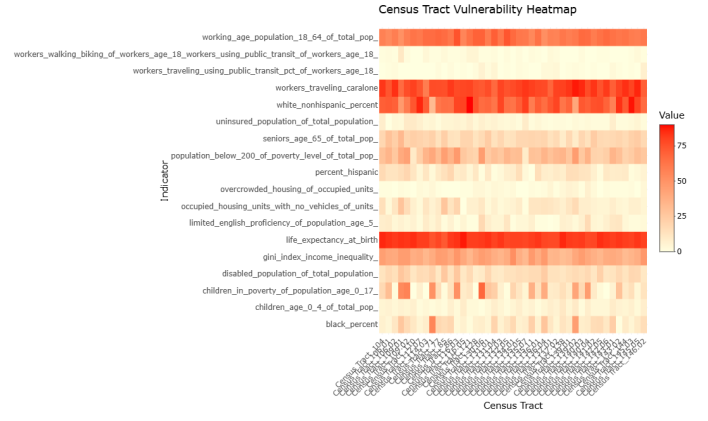
Figure 2: PCA plot with k-means clusters showing separation among census tract vulnerability profiles.

1.3 Clusters Comparison

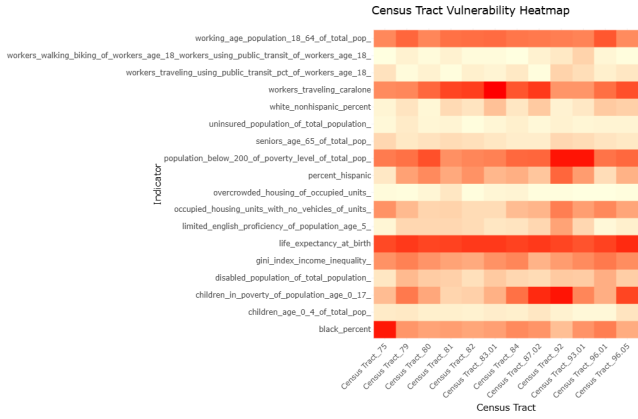
The Cluster 2 and Cluster 3 heatmap shows that Cluster 3 has the strongest concentration of high-risk indicators. Tracts in Cluster 3 have higher poverty, larger Black and Hispanic population shares, more limited English proficiency, more no-vehicle households, and more overcrowded housing. Cluster 2 appears more mixed and moderate, making it a transition group between low-risk and high-risk areas.



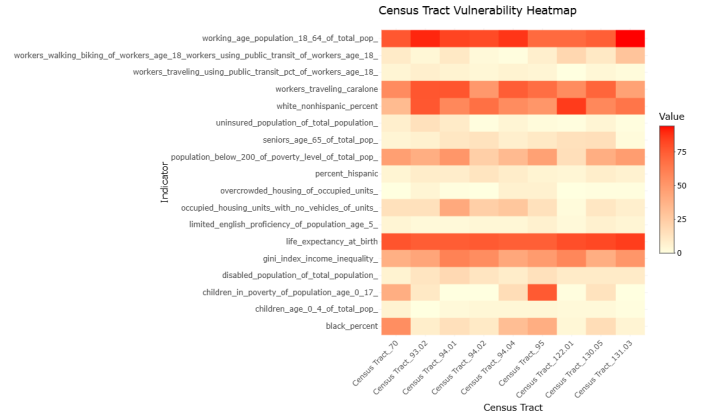
(a) Cluster 1 Heatmap



(b) Cluster 2 Heatmap



(c) Cluster 3 Heatmap



(d) Cluster 4 Heatmap

Figure 3: vulnerability heatmaps and PCA cluster separation.

1.4 Top Vulnerable Tracts

The top vulnerable tract list shows that the most at-risk neighborhoods are concentrated in Cluster 3 and, to a lesser extent, Cluster 4. These tracts have the highest average risk scores and share a common pattern of disadvantage: higher poverty, higher minority concentration, lower life expectancy, and greater housing and transportation barriers. Census Tracts 92, 93.01, 96.05, 87.02, 79, 96.01, 84, and 83.01 are among the strongest examples of this high-risk profile.

1.5 Indicator-Level Interpretation

The indicator pattern suggests that vulnerability is driven by a combination of social, economic, and environmental stressors. High poverty, low vehicle access, crowded housing, limited English proficiency, and high uninsured rates tend to appear together in the most vulnerable tracts. Lower life expectancy reinforces that these are not just income-poor neighborhoods but places where structural disadvantage affects health outcomes.

2 Cluster Characteristics and Vulnerability Profiles

Table 1: Vulnerability Cluster Profiles

Cluster	Size	Risk Score	Key Characteristics
1 (Low)	66 tracts	23.06 ± 1.34	Affluent White suburbs: 86% White, 77% life expectancy, 0.2% transit
2 (Medium)	43 tracts	25.91 ± 1.92	Moderate urban: 30% poverty, 14% transit, 82% life expectancy
3 (HIGH RISK)	12 tracts	31.69 ± 2.53	Severe deprivation: 64% poverty, 47% Black, 37% Hispanic, 33% LEP
4 (Mixed)	9 tracts	27.09 ± 2.26	Working poor elderly: 81% working age, 18% seniors, 38% poverty

2.1 Cluster 1: Affluent Suburbs (Lowest Risk)

Cluster 1 Profile

- **66 tracts** — stable, resource-rich areas requiring minimal intervention.
- **86% White non-Hispanic**, 77-year life expectancy.
- **0.35% no vehicle households**, 0.2% public transit usage.
- Baseline for healthy suburban communities.

2.2 Cluster 2: Moderate Urban (Medium Risk)

Cluster 2 Profile

- **43 tracts** — urban neighborhoods with targeted intervention needs.
- 30% poverty, 14% public transit dependence.
- Mixed demographics: 70% White, 82-year life expectancy.

2.3 Cluster 4: Working Poor Elderly

Cluster 4 Profile

- **9 tracts** — aging workforce with economic insecurity.
- Unique profile: 81% working-age but 38% poverty.
- 18% seniors, the highest share among clusters.
- 18% crowded housing, moderate transit dependence.

2.4 Cluster 3: Extreme High-Risk Core (Critical)

Cluster 3 Profile

- 12 tracts — immediate crisis intervention priority.
- Highest vulnerability across multiple indicators.
- 64% below poverty, versus 13% in Cluster 1.
- 47% Black + 37% Hispanic, versus 14% in Cluster 1.
- 33% limited English proficiency, about 8 times Cluster 1.
- 50% uninsured, versus 4% in Cluster 1.
- 47 Gini inequality, the highest among clusters.

3 Key Actionable Insights

Table 2: Priority Matrix

Priority	Cluster	Action Required
1: Immediate	Cluster 3 (12 tracts)	Extreme 64% poverty and 84% minority concentration. Language barriers compound healthcare access.
2: Urgent	Cluster 4 (9 tracts)	Working poor seniors need income and housing support. Prevent elderly poverty trap.
3: Monitor	Cluster 2 (43 tracts)	Growing transit dependence signals urban stress.
4: Baseline	Cluster 1 (66 tracts)	Affluent reference areas.

4 Geographic Targeting Strategy

IMMEDIATE: 12 Cluster 3 tracts
URGENT: 9 Cluster 4 tracts
MONITOR: 43 Cluster 2 tracts
BASELINE: 66 Cluster 1 tracts

Final Interpretation

Cluster 3 represents 12% of tracts but likely contains a much larger share of Rochester’s most vulnerable population. Targeted intervention here will likely produce the highest health impact per dollar spent.

Technical Notes

Report generated from PCA and k-means analysis of 23 socioeconomic and health indicators across Rochester census tracts. Analysis performed using an R Shiny dashboard with standardized vulnerability scoring.

5 Appendix

5.1 Census Tracts by Cluster

Table 3: Census Tracts Grouped by Cluster

Cluster	Census Tracts
1	Census Tract 141.04, 112.09, 118, 112.1, 126, 112.01, 101, 115.07, 125, 117.11, 142.02, 135.1, 134.02, 140.01, 114.01, 113.03, 105, 114.02, 122.02, 111, 117.05, 135.09, 130.04, 145.01, 132.08, 103, 116.01, 117.07, 115.05, 123.08, 146.01, 117.12, 108, 135.08, 78.02, 119.04, 123.01, 115.03, 112.03, 133, 102, 117.1, 117.08, 119.03, 115.04, 115.06, 124.02, 135.03, 113.04, 127, 123.04, 123.07, 117.09, 123.05, 130.03
2	Census Tract 109.01, 116.03, 109.02, 130.06, 139.02, 140.04, 71, 136.04, 85, 104, 143.02, 86, 135.11, 106.01, 140.03, 131.01, 134.01, 77, 135.05, 143.01, 136.01, 131.04, 145.05, 145.03, 106.02, 141.03, 139.01, 137.02, 137.01, 128, 142.06, 132.03, 144, 122.01, 142.04, 113.02, 132.07
3	Census Tract 92, 93.01, 96.05, 87.02, 79, 96.01, 84, 83.01, 80, 81, 82, 95
4	Census Tract 95, 94.01, 93.02, 94.04, 70, 130.05, 131.03, 94.02, 122.01